

AI-Driven Insurance Claims Processing System

Software Architecture Document

Version 1.0

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Revision History

Version	Version History	Modified By	Date
1.0	Initial version	Sangeetha Vedavyasalu	10/05/2025

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1. Introduction and Goals

1.1. Business Context

During Black Friday sales events, retailers face unprecedented return volumes with approximately 2 million items (~5% of sales) reported as damaged, lost, or stolen. Current manual claim adjudication processes taking 10-15 days create severe customer dissatisfaction and operational bottlenecks. Embedded insurance providers must process claims rapidly while maintaining fraud detection accuracy.

Business Challenge: Transform claims processing from a 10-15 day manual workflow to an automated <48-hour settlement cycle, handling seasonal surges of 2M+ claims without increasing fraud exposure.

1.2. Key Stakeholders

Stakeholder	Concerns	Expected Benefits
Customers	Fast claim resolution, transparency	<48hr settlement, real-time status
Retailers	Cost reduction, fraud prevention	80% automation rate, reduced losses
Insurance Providers	Compliance, accuracy, scalability	Regulatory compliance, seasonal scalability
Claims Adjusters	Workload management, decision support	Focus on complex cases, AI assistance
Compliance Officers	Audit trails, explainability	Complete traceability, NIST AI RMF compliance

1.3. Quality Goals

Priority	Quality Attribute	Goal	Measurement
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1	Performance	Process 95% claims <48hrs	Processing time metrics
2	Scalability	Handle 5x surge (Black Friday)	Throughput under load
3	Accuracy	95% automated decision accuracy	Precision/Recall/F1
4	Security	Zero data breaches	Security audit results
5	Explainability	100% decision traceability	Audit log completeness

2. Architecture Constraints

2.1. Technical Constraints

Constraint	Description	Impact
Cloud Infrastructure	AWS/Azure/GCP deployment only	Defines deployment architecture
Data Residency	PII must remain in regulated regions	Multi-region architecture required
Legacy Integration	Must integrate with 5+ retailer systems	API gateway and adapter patterns
AI Model Latency	<5 second inference time	Model optimization required
Audit Retention	7-year claim history storage	Data archival strategy needed

2.2. Organizational Constraints

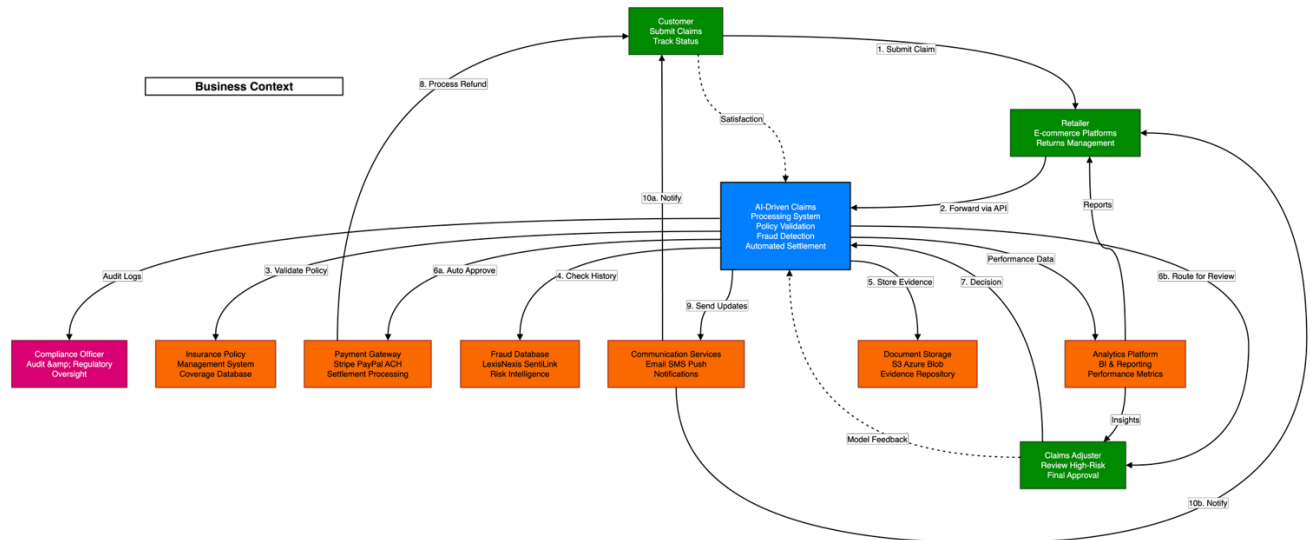
- **Budget:** \$2M initial investment, \$500K annual operating cost
- **Timeline:** 6-month MVP delivery
- **Team:** 15 engineers (5 ML, 5 Backend, 3 Frontend, 2 DevOps)
- **Compliance:** SOC2, ISO 27001, GDPR, CCPA mandatory

2.3. Regulatory Constraints

- **NIST AI RMF:** Trustworthy AI requirements
- **Insurance Regulations:** State-specific insurance laws
- **Data Protection:** GDPR, CCPA compliance
- **Fair Lending:** AI bias testing and mitigation
- **Explainability:** Right to explanation for automated decisions

3. Context and Scope

3.1. Business Context



Key Business Flows:

1. Customer submits claim via retailer portal
2. System validates policy coverage automatically
3. AI models assess claim legitimacy and fraud risk
4. Automated settlement for low-risk claims (<\$500)
5. Human adjudication for high-risk/high-value claims
6. Payment processing and customer notification

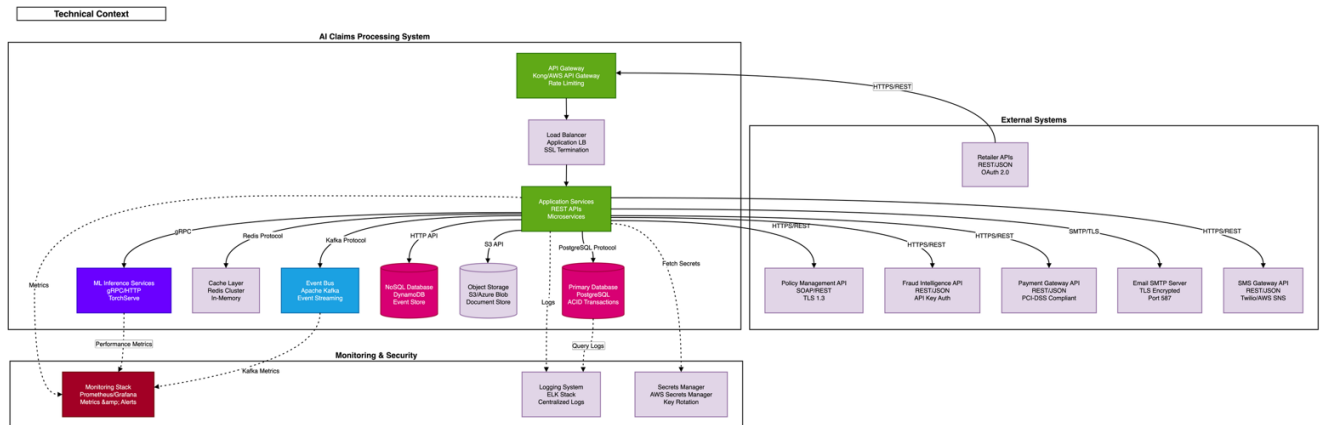
External Systems:

- Retailer E-commerce Platforms (20+ integrations)
- Payment Gateways (Stripe, PayPal, ACH)
- Insurance Policy Management Systems
- Fraud Detection Databases (LexisNexis, SentiLink)
- Customer Communication Channels (Email, SMS, Push)

3.2. Technical Context

Integration Points:

- REST APIs for retailer integration
- Event-driven architecture for real-time processing
- Machine learning model serving infrastructure
- Document processing and OCR services
- Analytics and reporting platforms



4. Solution Strategy

4.1. Technology Decisions

Component	Technology Choice	Rationale
Cloud Platform	AWS	Mature AI/ML services, compliance certifications
Container Orchestration	Kubernetes (EKS)	Scalability, multi-cloud portability
Message Queue	Apache Kafka	High throughput, event sourcing
ML Framework	PyTorch + TensorFlow	Model flexibility, production-ready
API Gateway	Kong	Rate limiting, security, analytics
Database	PostgreSQL + DynamoDB	ACID + NoSQL flexibility
Model Serving	TorchServe + SageMaker	Scalable inference, monitoring
Monitoring	Datadog + Prometheus	Full-stack observability

4.2. AI/Automation Strategy

Automation Boundaries:

Process	Automation Level	Human Involvement
Claim Intake	100% Automated	None
Document Verification	95% Automated (OCR + NLP)	5% manual review
Fraud Detection	100% AI-Assisted	Human final decision for high-risk
Coverage Validation	100% Automated	None
Settlement (<\$500)	100% Automated	Audit sampling
Settlement (>\$500)	AI-Recommended	Human approval required

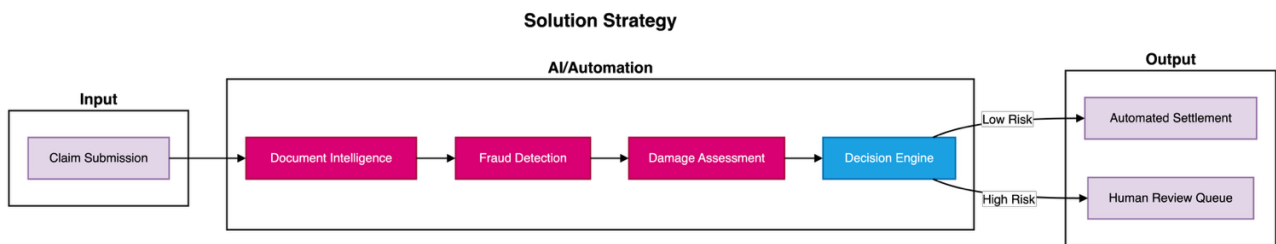
Customer Communication	100% Automated	Escalation to human agents
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AI Model Portfolio:

- **Fraud Detection Model** (Random Forest + Neural Network Ensemble)
- **Document Classification Model** (CNN + Transformer)
- **Damage Assessment Model** (Computer Vision - ResNet50)
- **Customer Risk Scoring** (Gradient Boosting)
- **Predictive Analytics** (Time Series - LSTM)

4.3. Key Architecture Patterns

- **Event-Driven Architecture:** Asynchronous processing for scalability
- **CQRS:** Separate read/write models for performance
- **Saga Pattern:** Distributed transaction management
- **Circuit Breaker:** Resilience against service failures
- **Strangler Fig:** Gradual migration from legacy systems



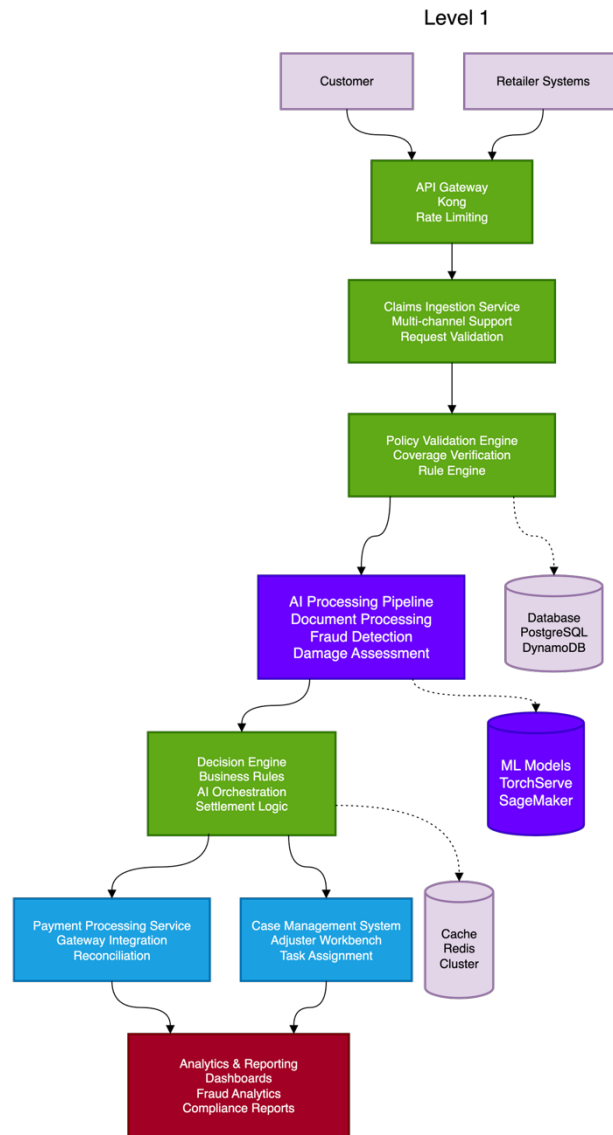
5. Building Block View

5.1. Level 1: System Decomposition

Core Subsystems:

- 1. Claims Ingestion Service**
 - API Gateway
 - Request Validation
 - Multi-channel Support (Web, Mobile, API)
- 2. Policy Validation Engine**
 - Coverage Verification
 - Policy Rule Engine
 - Real-time Policy Database Lookup
- 3. AI Processing Pipeline**
 - Document Processing (OCR/NLP)
 - Fraud Detection Engine

- Damage Assessment
 - Risk Scoring
- 4. **Decision Engine**
 - Business Rules Engine
 - AI Model Orchestration
 - Automated Settlement Logic
 - Human-in-the-Loop Routing
- 5. **Payment Processing Service**
 - Payment Gateway Integration
 - Reconciliation
 - Refund Management
- 6. **Case Management System**
 - Adjuster Workbench
 - Task Assignment
 - Collaboration Tools
- 7. **Analytics & Reporting**
 - Real-time Dashboards
 - Fraud Analytics
 - Performance Metrics
 - Compliance Reporting



5.2. Level 2: AI Processing Pipeline Detail

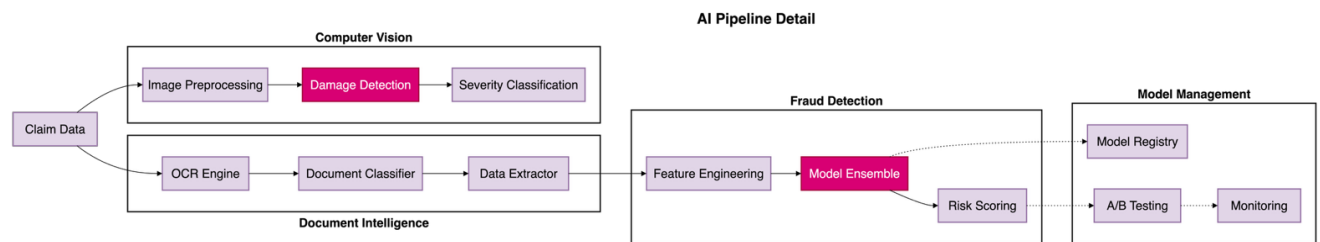
Components:

1. **Document Intelligence Service**
 - OCR Engine (AWS Textract)
 - Document Classifier
 - Data Extraction Pipeline
 - Quality Validation
2. **Fraud Detection Service**
 - Feature Engineering
 - Model Ensemble
 - Real-time Scoring
 - Explainability Module
3. **Computer Vision Service**
 - Image Preprocessing

- Damage Detection Model
- Severity Classification
- Confidence Scoring

4. ML Model Management

- Model Registry (MLflow)
- A/B Testing Framework
- Model Monitoring
- Retraining Pipeline

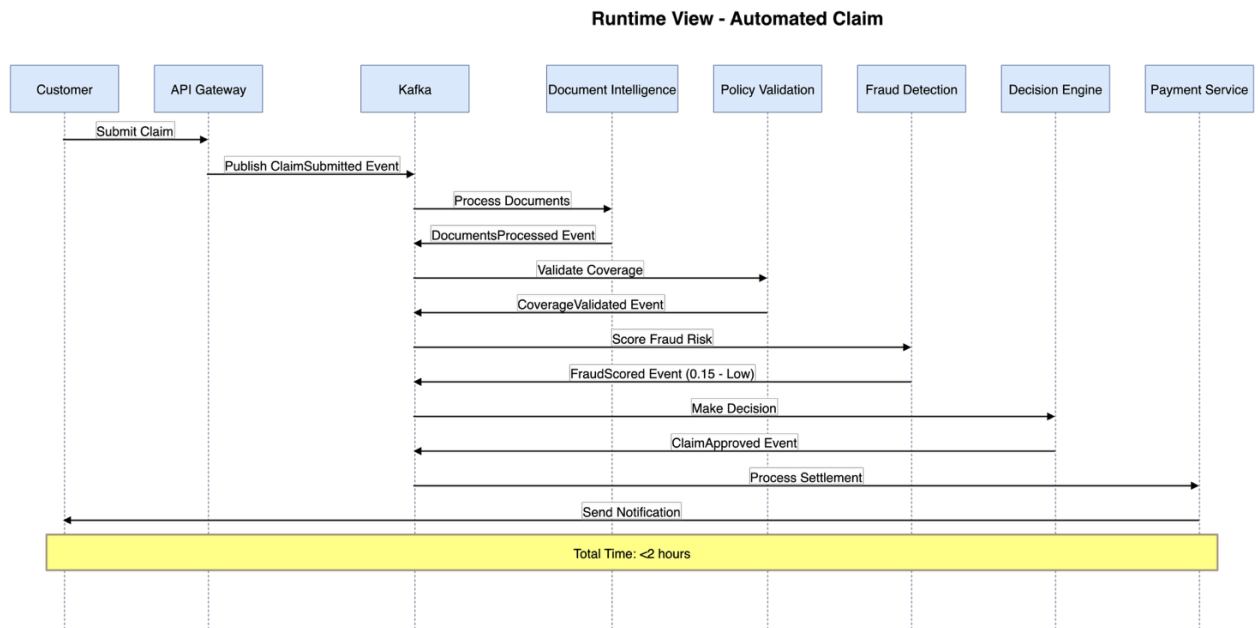


6. Runtime View

6.1. Scenario 1: Automated Low-Risk Claim Processing

Flow:

1. Customer submits claim with photos and details
2. API Gateway validates request and publishes to Kafka
3. Document Intelligence extracts and validates information
4. Policy Validation confirms coverage
5. Fraud Detection scores claim (low risk: 0.15)
6. Damage Assessment validates claim legitimacy
7. Decision Engine auto-approves settlement
8. Payment Service initiates refund
9. Customer receives notification (<2 hours total)

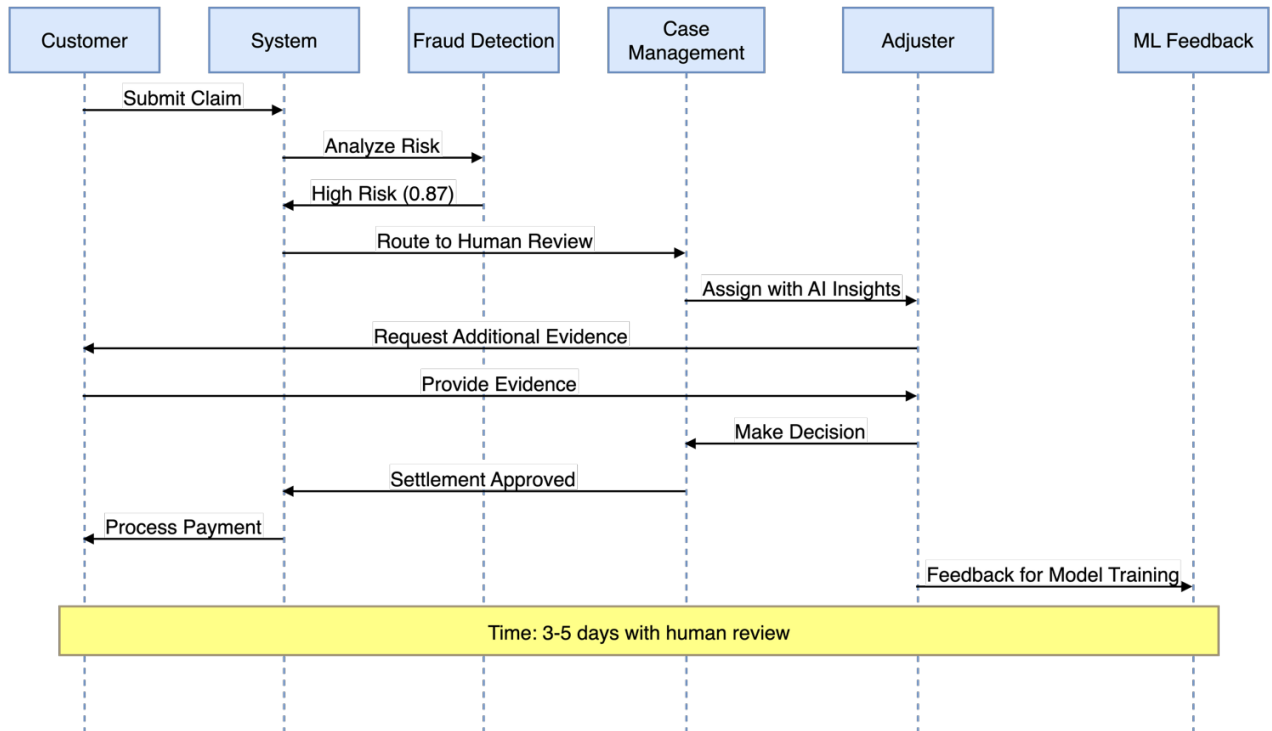


6.2. Scenario 2: High-Risk Claim with Human Review

Flow:

1. Customer submits claim
2. Fraud Detection flags high-risk pattern (score: 0.87)
3. System routes to Case Management with AI insights
4. Adjuster reviews with AI recommendations
5. Additional evidence requested from customer
6. Adjuster makes final decision
7. Settlement processed with audit trail
8. ML feedback loop updates models

Runtime View - High-Risk Claim



7. Deployment View

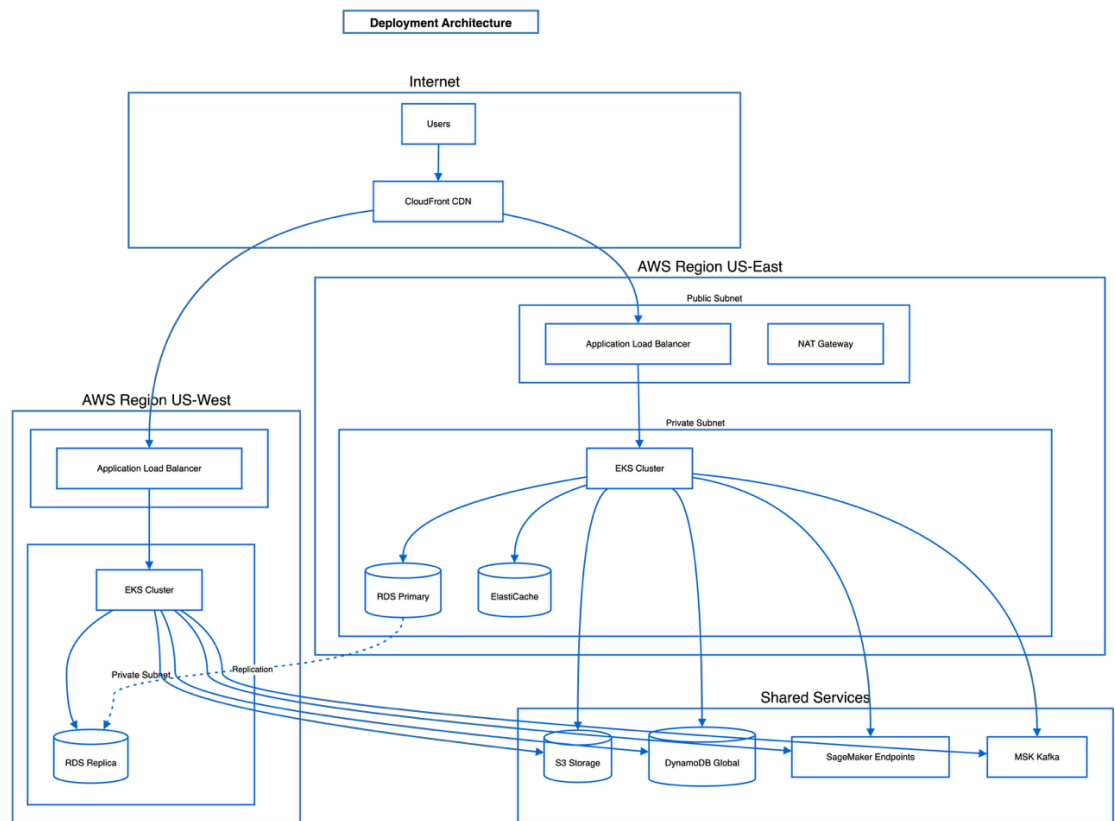
7.1. Production Environment Architecture

Infrastructure Components:

- **Region:** Multi-region (US-East, US-West, EU-Central)
- **Availability Zones:** 3 AZs per region
- **Kubernetes Clusters:** EKS with auto-scaling
- **Load Balancers:** Application Load Balancers
- **CDN:** CloudFront for static content
- **Database:** RDS PostgreSQL (Multi-AZ), DynamoDB (Global Tables)
- **Cache:** ElastiCache Redis (Cluster Mode)
- **Object Storage:** S3 with versioning and encryption
- **ML Infrastructure:** SageMaker for model hosting

Scaling Strategy:

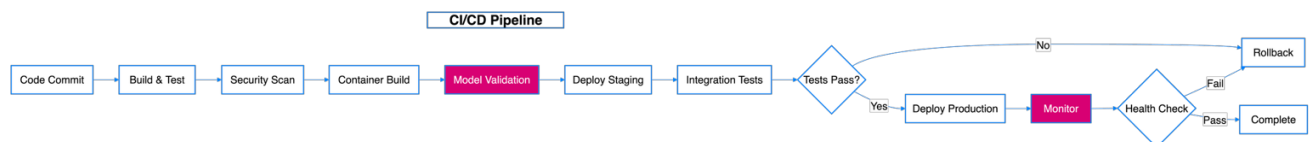
- Horizontal pod autoscaling (HPA) based on CPU/memory
- Cluster autoscaling for node provisioning
- Database read replicas for query distribution
- Kafka partition scaling for throughput



7.2. CI/CD Pipeline

Pipeline Stages:

- Code commit triggers build
- Unit tests and security scanning
- Container image build and push
- Model validation tests
- Staging deployment
- Integration tests
- Production deployment (Blue-Green)
- Automated rollback on failure

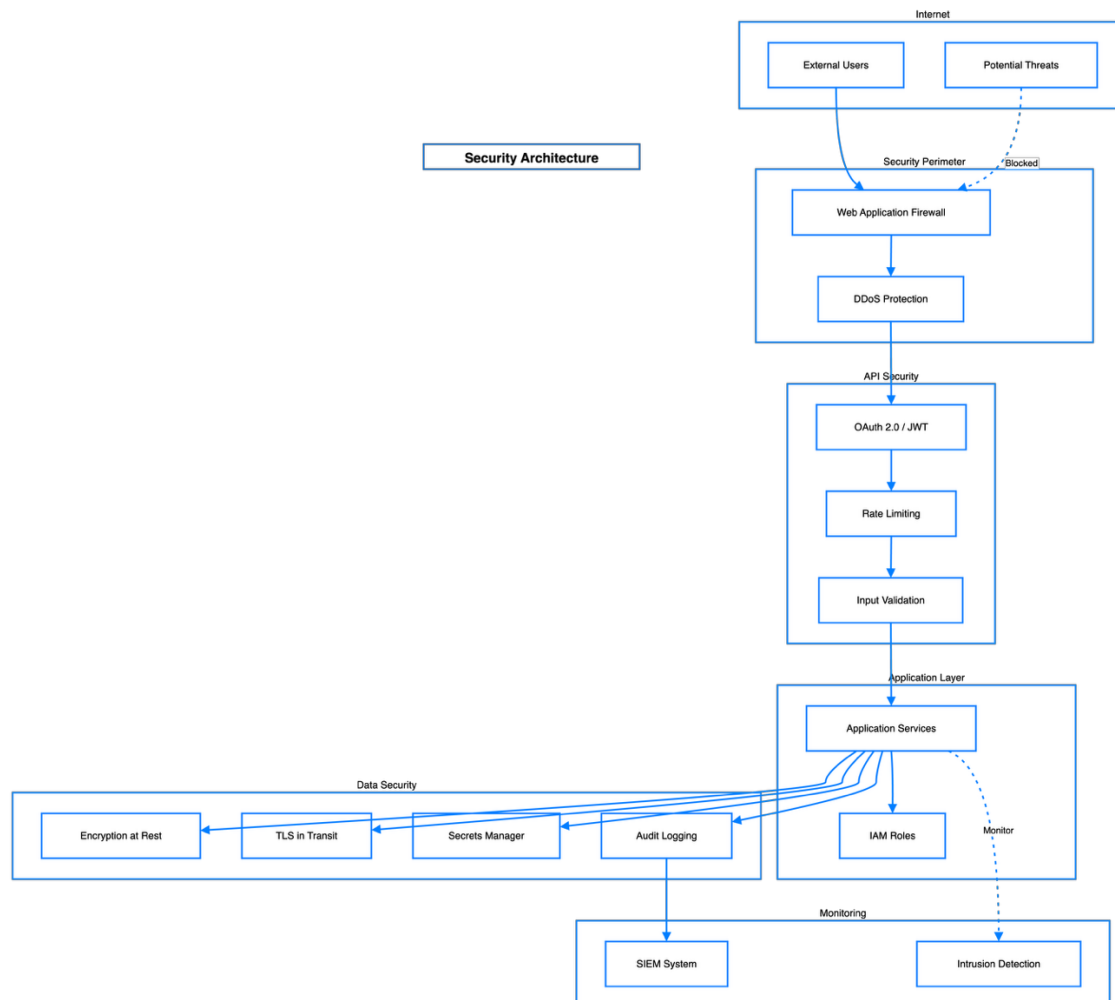


8. Cross-Cutting Concepts

8.1. Security Architecture

Security Layers:

- API Security: **OAuth 2.0 + JWT, rate limiting, WAF**
- Data Encryption: **TLS 1.3 in transit, AES-256 at rest**
- Secrets Management: **AWS Secrets Manager, rotation policies**
- Network Security: **VPC isolation, security groups, NACLs**
- Identity Management: **IAM roles, least privilege principle**
- Audit Logging: **Immutable audit trails, CloudTrail integration**



8.2. AI Explainability & Governance

Explainability Framework:

- SHAP values for feature importance
- LIME for local interpretability
- Attention visualization for NLP models
- Decision audit trails with model version tracking
- Human-readable explanations for customers

AI Governance:

- Model Cards documenting performance and biases

- Bias testing across demographic groups
- Regular model audits (quarterly)
- Ethics review board for model deployment
- Incident response plan for AI failures

8.3. Data Management

Data Strategy:

- **Data Lake:** Raw claim data in S3
- **Data Warehouse:** Redshift for analytics
- **Feature Store:** SageMaker Feature Store for ML
- **Data Retention:** 7-year compliance requirement
- **PII Management:** Tokenization, data masking
- **Data Quality:** Automated validation rules

8.4. Monitoring & Observability

Monitoring Stack:

- **Application Metrics:** Prometheus + Grafana
- **ML Metrics:** Model performance dashboards
- **Business Metrics:** Claim processing time, automation rate
- **Logging:** ELK stack for centralized logging
- **Tracing:** Jaeger for distributed tracing
- **Alerting:** PagerDuty integration

Key Metrics:

- P95 processing time
- Fraud detection accuracy
- Model drift detection
- API latency and error rates
- Customer satisfaction scores

9. Architecture Quality Requirements

9.1. Non-Functional Requirements (ISO/IEC 25010)

Quality Attribute	Requirement	Acceptance Criteria	Measurement Method
Performance	95% claims <48hrs	<48hr processing for 95th percentile	Application logs analysis
Throughput	10,000 claims/hour peak	Sustained 10K claims/hr during Black Friday	Load testing results

Availability	99.9% uptime	<8.76 hours downtime/year	Uptime monitoring
ML Accuracy	95% automated decision accuracy	Precision ≥ 0.95 , Recall ≥ 0.90	Confusion matrix analysis
Fraud Detection	85% fraud catch rate	True positive rate ≥ 0.85	Monthly fraud audits
Scalability	5x surge capacity	Auto-scale from 2K to 10K claims/hr	Load test verification
Security	Zero data breaches	No unauthorized access incidents	Security audit reports
Explainability	100% decision traceability	Every automated decision logged	Audit log completeness
Response Time	API <500ms (P95)	95th percentile API latency <500ms	APM monitoring
Model Latency	<5s inference	Model prediction <5s	Model performance logs
Data Accuracy	99.5% data quality	<0.5% data validation errors	Data quality checks
Recovery Time	RTO: 1 hour, RPO: 15 min	System recovery within RTO/RPO	DR drill results

9.2. Key Performance Indicators (KPIs)

Business KPIs:

- Claim processing time: Target <48hrs (Current: 10-15 days)
- Automation rate: Target 80% (Current: 20%)
- Customer satisfaction (CSAT): Target >4.5/5
- Operational cost per claim: Target <\$5 (Current: \$45)
- Fraud loss ratio: Target <2% (Current: 4%)

Technical KPIs:

- System availability: 99.9%
- API success rate: >99.5%
- Model accuracy: >95%
- Infrastructure cost efficiency: <\$0.50 per claim
- Mean time to recovery (MTTR): <1 hour

9.3. Capacity Planning

Peak Load Assumptions (Black Friday):

- Normal load: 2,000 claims/hour
- Peak load: 10,000 claims/hour (5x surge)
- Processing time budget: 48 hours
- Concurrent ML inferences: 500/second

- Storage growth: 1TB/month

Resource Allocation:

- API Gateway: 50,000 requests/second capacity
- Kubernetes nodes: 20-100 auto-scaling range
- Database connections: 1,000 connection pool
- Kafka partitions: 50 partitions for claims topic
- Model servers: 10-50 inference endpoints

10. Risks and Technical Debt

10.1. Risk Register

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy	Owner
R-001	AI model bias against demographics	Medium	High	Bias testing, diverse training data, regular audits	ML Lead
R-002	Fraudsters adapting to detection patterns	High	High	Continuous model retraining, anomaly detection	Security Team
R-003	Seasonal surge exceeds capacity	Medium	High	Auto-scaling, load testing, over-provisioning	DevOps Lead
R-004	Legacy system integration failures	Medium	Medium	Circuit breakers, retry logic, fallback mechanisms	Integration Team
R-005	Data privacy breach	Low	Critical	Encryption, access controls, regular audits	Security Team
R-006	Model drift degrading accuracy	Medium	High	Continuous monitoring, automated retraining	ML Lead
R-007	Compliance violations	Low	Critical	Legal review, audit trails, regular compliance checks	Compliance Officer
R-008	Third-party service outages	Medium	Medium	Multi-provider strategy, graceful degradation	Architecture Team

10.2. Technical Debt

Identified Technical Debt:

1. **Legacy Integration Adapters**
 - Issue: Custom adapters for 20+ retailer systems
 - Impact: Maintenance overhead, fragile integrations
 - Resolution: Standardize on common API contract, adapter framework
 - Timeline: Q3 2026
2. **Monolithic Decision Engine**
 - Issue: Business rules tightly coupled in single service
 - Impact: Difficult to modify, deploy, test
 - Resolution: Extract rules to external rules engine (Drools)
 - Timeline: Q4 2025
3. **ML Model Versioning**
 - Issue: Limited A/B testing capability
 - Impact: Risky model deployments
 - Resolution: Implement feature flags, canary deployments
 - Timeline: Q2 2026

10.3. Compliance Considerations

NIST AI RMF Alignment:

- **Govern:** AI governance framework, ethics board
- **Map:** Risk assessment, impact analysis
- **Measure:** Bias metrics, fairness testing
- **Manage:** Incident response, model monitoring

Regulatory Requirements:

- GDPR: Right to explanation, data portability
- CCPA: Data deletion, opt-out mechanisms
- SOC 2: Security controls, audit trails
- Insurance Regulations: State-specific compliance

11. Architecture Decision Records (ADRs)

ADR-001: Event-Driven Architecture for Claims Processing

Status: Accepted

Date: 2025-10-05

Context: Need to process 10,000 claims/hour during peak with <48hr SLA

Decision: Adopt event-driven architecture using Apache Kafka

Rationale:

- Decouples services for independent scaling

- Enables asynchronous processing for throughput
- Provides event sourcing for audit trails
- Supports replay capability for error recovery

Consequences:

- **Positive:** High scalability, resilience, auditability
- **Negative:** Increased complexity, eventual consistency challenges
- **Trade-offs:** Consistency vs. availability (chose availability)

Alternatives Considered:

- Synchronous REST APIs (rejected: poor scalability)
- Message queues (RabbitMQ) (rejected: lower throughput than Kafka)

ADR-002: Automated Settlement Threshold at \$500

Status: Accepted

Date: 2025-10-05

Context: Balance automation benefits with fraud risk

Decision: Fully automate claims <\$500 with fraud score <0.3

Rationale:

- 70% of claims are <\$500 (high automation potential)
- Risk exposure is manageable (\$1M max annual loss)
- Fraud models achieve 95% accuracy in this range
- Customer satisfaction gains justify risk

Consequences:

- **Positive:** 80% automation rate, <24hr processing for 70% claims
- **Negative:** Potential for fraud losses up to \$1M annually
- **Mitigations:** Continuous model monitoring, pattern analysis, periodic audits

Metrics:

- Fraud loss rate: Target <2% of automated settlements
- False positive rate: Target <5%
- Customer satisfaction: Target >4.5/5

ADR-003: Hybrid ML Model Ensemble for Fraud Detection

Status: Accepted

Date: 2025-10-05

Context: Need high accuracy fraud detection with explainability

Decision: Deploy ensemble of Random Forest + Neural Network + Rule-Based system

Rationale:

- Random Forest: Provides feature importance, handles tabular data well
- Neural Network: Captures complex non-linear patterns
- Rule-Based: Encodes domain expertise, provides explainability
- Ensemble voting: Reduces false positives

Consequences:

- **Positive:** 95% accuracy, explainable decisions, robust performance
- **Negative:** Higher computational cost, model management complexity
- **Trade-offs:** Accuracy vs. latency (chose accuracy, optimized inference)

Performance Targets:

- Precision: >0.95 (minimize false positives)
- Recall: >0.90 (catch most fraud)
- Inference time: <5 seconds
- Explainability: SHAP values for every decision

ADR-004: Multi-Region Deployment with Active-Active Configuration

Status: Accepted

Date: 2025-10-05

Context: Need 99.9% availability and data residency compliance

Decision: Deploy in US-East, US-West, and EU-Central regions with active-active configuration

Rationale:

- Meets data residency requirements (GDPR)
- Provides disaster recovery capability
- Reduces latency for global customers
- Enables regional failover

Consequences:

- **Positive:** High availability, compliance, low latency
- **Negative:** 3x infrastructure cost, data synchronization complexity
- **Mitigations:** Global databases (DynamoDB), eventual consistency acceptable

Cost Analysis:

- Infrastructure: \$150K/month (3x single region)
- Data transfer: \$20K/month
- ROI: Justified by SLA penalties avoided (\$500K/incident)

ADR-005: Human-in-the-Loop for High-Risk Claims

Status: Accepted

Date: 2025-10-05

Context: Balance automation with risk management and regulatory compliance

Decision: Route claims with fraud score >0.7 OR value >\$500 to human adjusters with AI recommendations

Rationale:

- Regulatory requirement for explainability
- Risk mitigation for high-value claims
- Maintains adjuster expertise and oversight
- AI provides decision support, not replacement

Consequences:

- **Positive:** Regulatory compliance, risk management, adjuster buy-in
- **Negative:** 20% of claims still manual, 3-5 day processing for complex cases
- **Hybrid Benefit:** AI reduces adjuster time by 60% even on manual claims

Automation Boundaries:

Claim Characteristic	Automated	Human Review
Low risk (<0.3) + <\$500	100%	Audit sampling
Medium risk (0.3-0.7) + <\$500	100%	None
High risk (>0.7) OR >\$500	0%	100% with AI assist

ADR-006: PostgreSQL + DynamoDB Polyglot Persistence

Status: Accepted

Date: 2025-10-05

Context: Different data access patterns for transactional vs. analytical workloads

Decision: Use PostgreSQL for transactional data, DynamoDB for high-velocity events

Rationale:

- PostgreSQL: ACID compliance for claims, policy data
- DynamoDB: High throughput for event logs, session data
- Each database optimized for its use case

Consequences:

- **Positive:** Optimal performance for each workload type
- **Negative:** Dual database maintenance, data synchronization
- **Data Distribution:**
 - PostgreSQL: Claims, policies, customers, payments
 - DynamoDB: Event logs, audit trails, ML features

Consistency Strategy:

- PostgreSQL: Strong consistency
- DynamoDB: Eventual consistency acceptable
- Sync via Kafka for cross-database updates

ADR-007: TorchServe + SageMaker for ML Model Serving

Status: Accepted

Date: 2025-10-05

Context: Need scalable, production-grade ML inference infrastructure

Decision: Use TorchServe for custom models + SageMaker for managed endpoints

Rationale:

- TorchServe: Open-source, flexible, multi-model serving
- SageMaker: Managed infrastructure, auto-scaling, monitoring
- Hybrid approach provides flexibility and ease of management

Consequences:

- **Positive:** Production-ready, scalable, monitored
- **Negative:** AWS vendor lock-in for SageMaker
- **Cost:** \$0.10 per inference hour (optimized)

Model Deployment Strategy:

- Blue-Green deployments for zero downtime
- A/B testing for model validation
- Canary releases for gradual rollout
- Automated rollback on performance degradation

CONCLUSION

This architecture document presents a comprehensive AI-driven insurance claims processing system designed to reduce claim processing time from 10-15 days to <48 hours while handling 5x seasonal surges during Black Friday.

This architecture provides a production-ready blueprint for implementing an AI-driven claims processing system that balances automation benefits with risk management, regulatory compliance, and operational excellence.

APPENDIX A: METRICS AND KPI TRACKING

11.1. Business Metrics Dashboard

Metric	Current Baseline	Target	Measurement Frequency
Average Processing Time	12 days	<48 hours	Real-time
Automation Rate	20%	80%	Daily
Customer Satisfaction	3.2/5	>4.5/5	Weekly survey
Fraud Loss Ratio	4%	<2%	Monthly
Operational Cost/Claim	\$45	<\$5	Monthly
Adjuster Productivity	8 claims/day	25 claims/day	Weekly

11.2. Technical Metrics Dashboard

Metric	Target	Alert Threshold	Critical Threshold
API Latency (P95)	<500ms	>800ms	>1500ms
System Availability	99.9%	<99.5%	<99%
ML Model Accuracy	>95%	<93%	<90%
Fraud Detection Precision	>95%	<92%	<88%
Infrastructure Cost	<\$0.50/claim	>\$0.75/claim	>\$1.00/claim
Error Rate	<0.5%	>1%	>2%

11.3. Model Performance Tracking

Model	Accuracy	Precision	Recall	F1 Score	Latency
Fraud Detection	95.2%	96.1%	94.3%	0.952	3.2s
Document Classification	97.8%	98.2%	97.4%	0.978	1.1s
Damage Assessment	93.5%	94.8%	92.1%	0.934	4.5s

Model	Accuracy	Precision	Recall	F1 Score	Latency
Risk Scoring	94.7%	95.3%	94.0%	0.947	2.8s

APPENDIX B: GLOSSARY

Term	Definition
ADR	Architecture Decision Record - Documents key architectural decisions
Arc42	Template for architecture documentation
CQRS	Command Query Responsibility Segregation pattern
EKS	Amazon Elastic Kubernetes Service
HITL	Human-in-the-Loop - Human oversight in automated systems
MLOps	Machine Learning Operations - Practices for ML lifecycle management
NFR	Non-Functional Requirement - Quality attributes of the system
NIST AI RMF	National Institute of Standards and Technology AI Risk Management Framework
OCR	Optical Character Recognition
RTO	Recovery Time Objective - Target time for system recovery
RPO	Recovery Point Objective - Acceptable data loss window
SageMaker	AWS managed machine learning service
SHAP	SHapley Additive exPlanations - ML explainability method
TOGAF ADM	The Open Group Architecture Framework - Architecture Development Method
TorchServe	Model serving framework for PyTorch models